

Blitz presentation: ACL 2026 paper

More Aligned, Less Diverse?
Analyzing the Grammar and Lexicon
of Two Generations of LLMs

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Overview

- Compared human news (NYT) paragraphs with LLM text
- Contrasted LLM output from 2023 vs. 2025 (instruction-tuned)
- Parsed all texts with ERG, storing only top-ranked analysis
- Measured diversity of lexical types and constructions
- Found less diversity in 2025 LLMs (lex and rules) than for 2023
- 2025 LLM text sentences are longer but easier for ERG to parse
- 2025 LLMs have far fewer fragments and short sentences (1-15)

Datasets

Dataset	# Sent. in dataset	Model size	Training tokens	Data sources
Qwen 2.5	37,825	14B	18T	Not disclosed
	26,498	32B	18T	
	34,892	72B	18T	
LlaMa 3.3	39,306	70B	15T+	Not disclosed
Mistral v.0.3	33,840	7B	1T	Not disclosed
GPT-4o	50,544	Not disclosed	13T	Not disclosed
Original NYT	26,102	N/A	N/A	New York Times

Prompts for LLM output

```
system_prompt: "You are a professional  
journalist specializing in writing news.  
Follow the given structure."
```

```
user_prompt: "You will write a news lead  
paragraph using the inputs below."
```

```
Inputs
```

```
Headline: {headline}
```

```
LeadThreeWords: {lead_three_words}
```

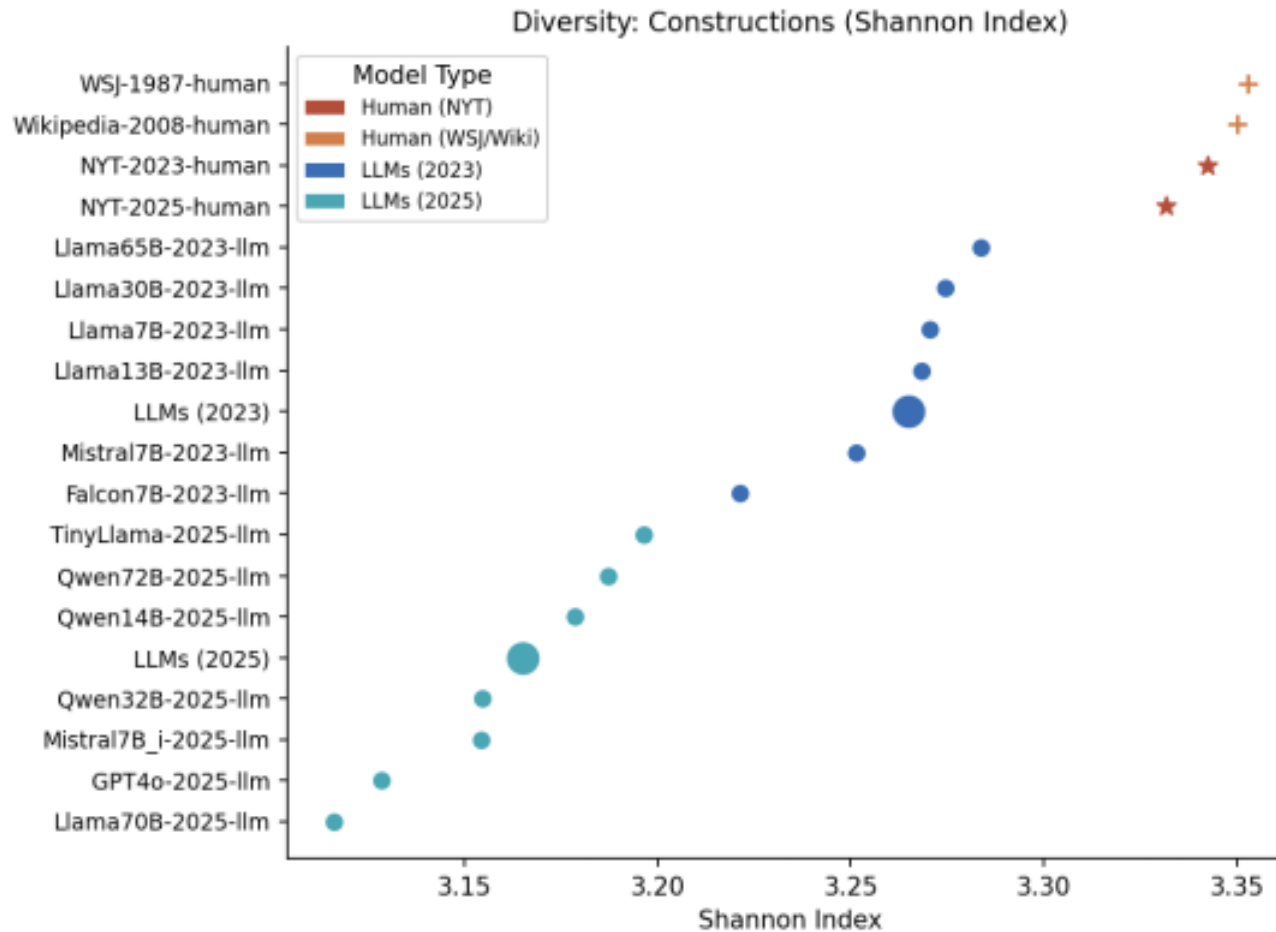
```
Requirements - Mandatory
```

```
Write one paragraph of several sentences (  
more than one, e.g. two-three (2-3)); no  
title, no bullets.
```

```
Output format: the paragraph only, no  
preamble or labels."
```

Diversity in constructions

Human text more diverse than all LLMs



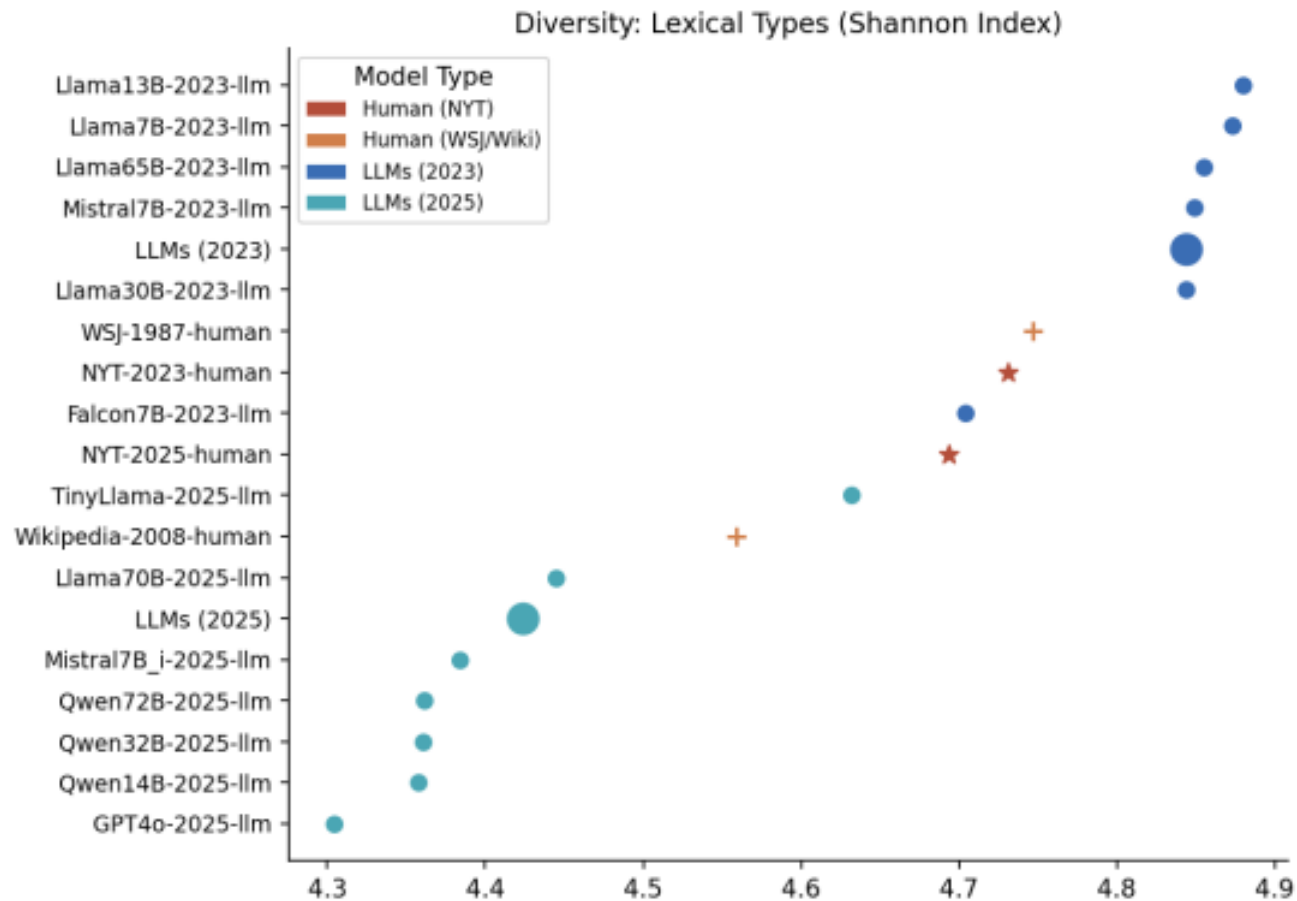
Constructions: LLM 2023 vs. 2025

2025 less diverse than 2023

Construction	Year	Example Sentence
Nominal head + preceding adjunct	2025	Phyllis Dalton, the Oscar-winning costume designer known for her meticulous work on historical epics, passed away at the age of 99.
Subordinate pred phrase from participial VP	2025	The Learjet was carrying a pediatric patient when it crashed, killing all on board.
Head + following scope adjunct	2025	Speaker Kevin McCarthy began the final day before the shutdown, facing dim prospects of passing a funding measure.
Bare NP	2025	Tariff strategy could harm economic growth and job creation.
Adjective–Headed Normal Construction	2023	“I am concerned about how they are using [the law],” Cardin said.
Bare NP	2023	Senator Ben Cardin told Al-Monitor that he will oppose the release of the remaining \$650 million in military aid to Egypt.
Subordinated Predicative VP (Participial)	2023	It was 40 degrees outside, but people were lined up wearing shorts and flip-flops.
Head–Adjective Small Clause (Predicative)	2023	When the Supreme Court returns on Monday from its summer recess, it will be without Justice Antonin Scalia.

Diversity in lexical types

2025 less diverse than human, 2023 most diverse



Human vs. LLM lexical types

Lex types	Preferred by
Adjective (intersective, non-comparative)	LLM
Transitive verb with NP complement	LLM
Count noun (lexical)	LLM
Count noun with non-specific reference	LLM
Proper noun (generic)	Human
Adj (intersective, coordinated, generic)	Human
Verb (PP + finite CP complement, implicative)	Human
Proper noun (referential)	Human

Lexical types: LLM 2023 vs. 2025

Lex types	Year	Example Sentence
Adjective (intersective, non-comparative)	2025	A look back at her career reveals a dedication to authenticity and detail that brought eras long past vividly to life on screen.
Transitive verb with NP complement	2025	Critics argue this action undermines collective bargaining rights and could lead to increased tensions with federal employee unions.
Count noun (lexical)	2025	A man was wounded and a nine-year-old boy was fatally shot during an incident in Newark, according to officials.
Count noun with non-specific reference	2025	The unrest has displaced thousands and raised concerns over the control of vital resources such as cobalt and copper.
Utterance particle	2023	I don't see any signs that inventories are excessive.
Pronoun (personal, first person singular, it)	2023	"It's not acceptable for a democracy."
Pronoun (personal, second person, you)	2023	"If you ran for president," he wondered, "would you be able to win in Iowa?"
Pronoun (personal, first person plural, we)	2023	There was a time when we could blame our problems on Bush, but no more.